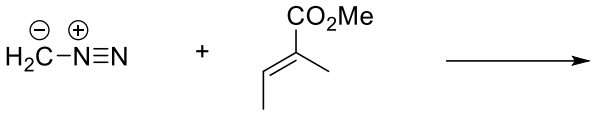
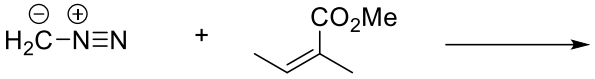
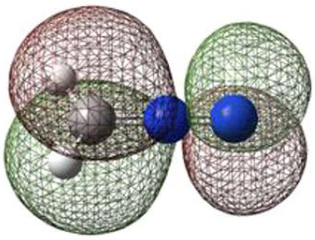
Question 1. Predict the products of the cycloaddition reactions of diazomethane.

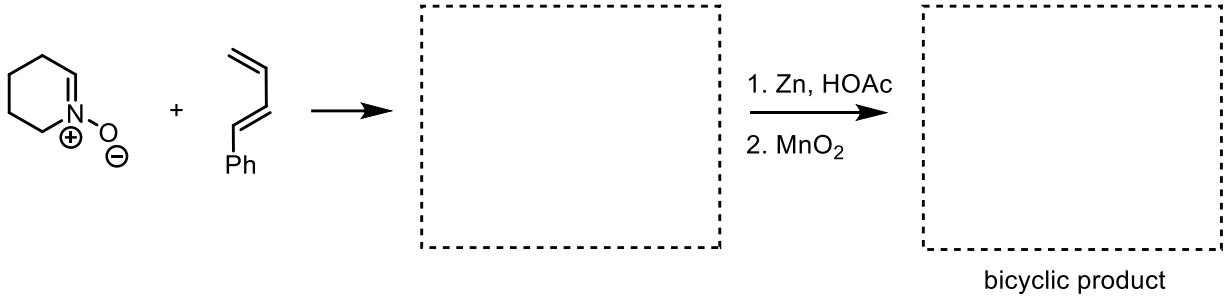






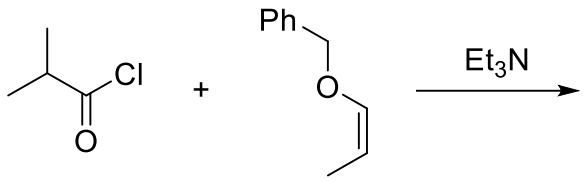
HOMO of diazomethane

Question 2. Propose the structures of products for the following reaction.

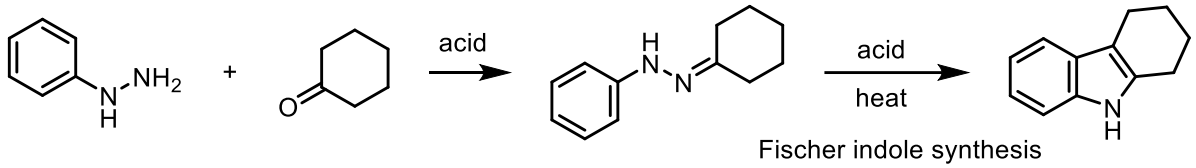


Explain why in the first step, the [4+3] cycloaddition does not take place.

Question 3. Predict the product of the flowing reaction and explain the regio- and stereo- selectivity of the reaction.



Question 4. Proposed the stepwise mechanism for Fischer indole synthesis.



Fischer indole synthesis

A couple of tautomerization steps are involved in Fischer indole synthesis. Are these tautomerization steps acid catalysed, or are they [1,n]-sigmatropic hydrogen shifts?